

Have fun!

1. (Algebra Comp Fall 2021 Rings) Let R be a commutative ring with unity. An element $a \in R$ is called *nilpotent* if $a^n = 0$ for some $n \in \mathbb{N}$ that can depend on a . Let $N \subseteq R$ be the subset of all nilpotent elements of R .
 - (a) Prove that N is an ideal of R .
 - (b) Prove that the quotient ring R/N contains no nonzero nilpotent elements.
2. Section 7.2 Problem 17, 18, 24 and 26abc (for part a, you may assume the map is a homomorphism and simply focus on the kernel)